

CivicTrack System

Phase 2: Requirements Specification

Analysis Stage, Detailed System
Blueprint

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1.Non-Functional Requirements

1.1. Reliability

- The system shall have minimal downtime.
- The system shall ensure that reports and user information are not lost in the event of a failure.
- The system shall recover after crashes.

1.2. Availability

- The system shall be available 24/7.
- The system shall have minimal service interruptions.

1.3. Security

- The system shall require users to log in using secure authentication (username and password).
- The system shall protect and encrypt user information.
- The system shall allow only authorized users (Admin, Staff) to access sensitive information.

1.4. Maintainability

- The system shall be easy for developers to update and fix.
- The system shall have well-organized and documented code.
- The system shall make it easy to find and fix bugs.

1.5. Portability

- The system shall be accessible on different devices (mobile and desktop).
- The system shall support different operating systems.

1.6. Performance

- The system shall support multiple users at once without experiencing slowdowns.
- The system shall store and send reports immediately.

2.Funtional Requirements Structure

2.1. Input

- The system shall allow citizens to submit service issues online.
- The system shall allow the user to enter issue details.
- The system shall allow the user to select a category, upload images as proof, and provide a location.

2.2. Process

- The system shall automatically assign reports to the correct municipal department.
- The system shall issue out the reference number for each report.
- The system shall allow municipal administrators to view reported issues.
- The system shall allow municipal administrators to assign tasks to technicians.
- The system shall allow administrators and staff to update issue status.
- The system shall allow administrators to escalate unresolved issues.

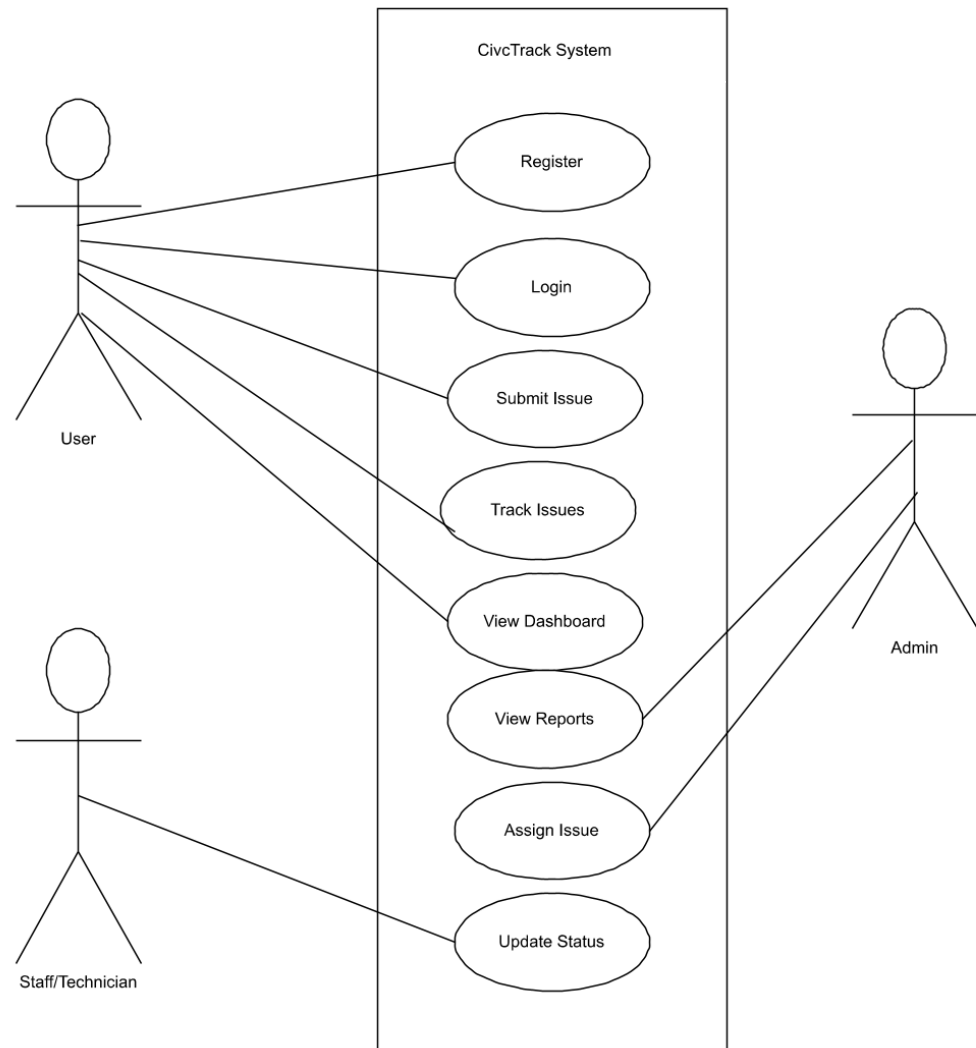
2.3. Output

- The system shall allow the user to track the status of their issues.
- The system shall allow the user to view progress via a dashboard.
- The system shall send notifications when issue is received, status changes and issue is resolved.
- The system shall display issues on a map.
- The system shall use color-coding to indicate status.
- The system shall show statistics like resolution time.

3. UML Diagrams Required

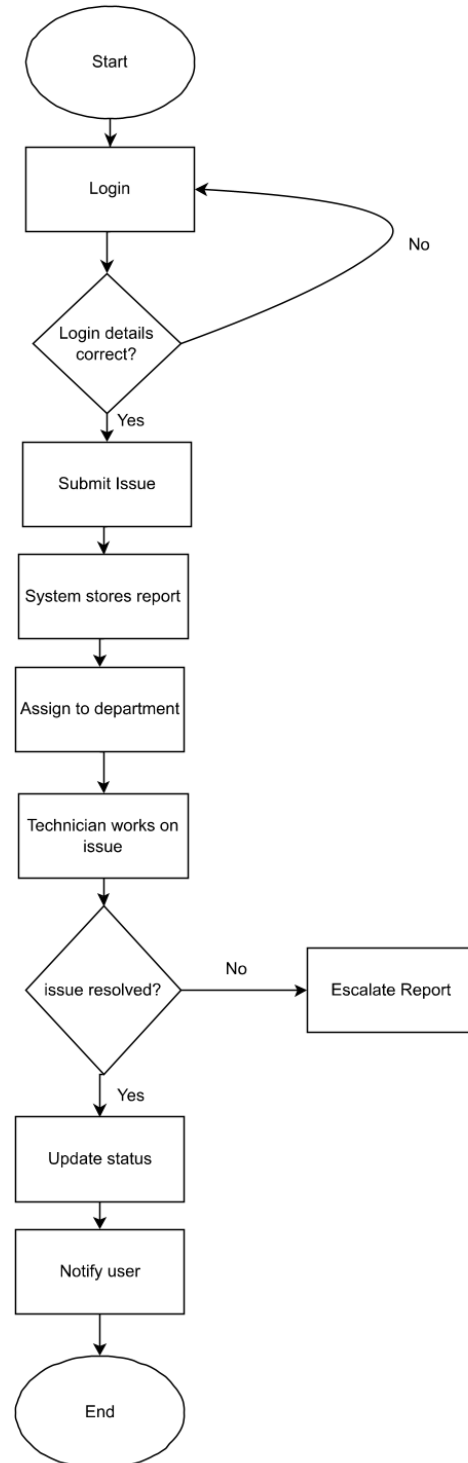
3.1. Use Case Diagram

USE CASE DIAGRAM

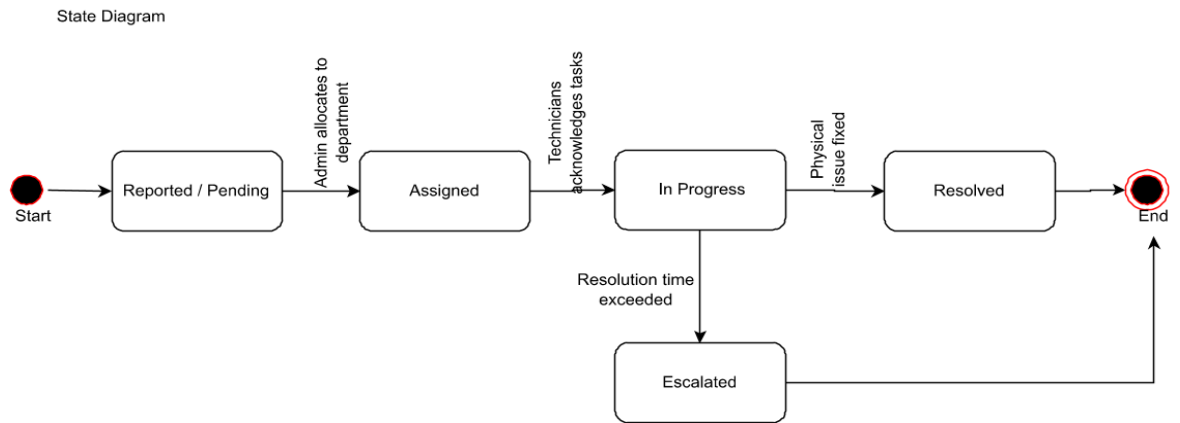


3.2. Activity Diagram

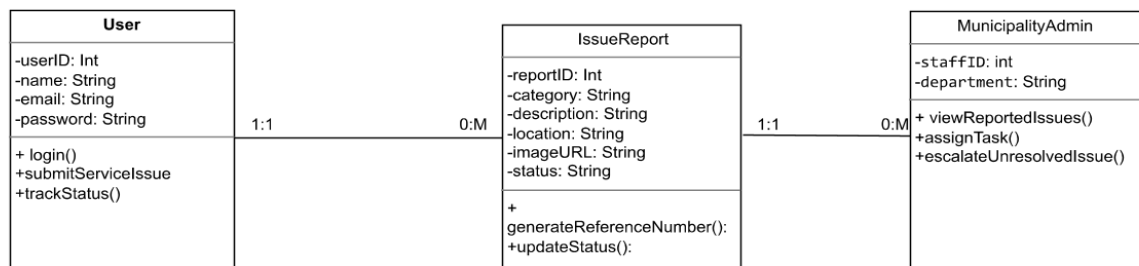
Activity Diagram



3.3. State Diagram

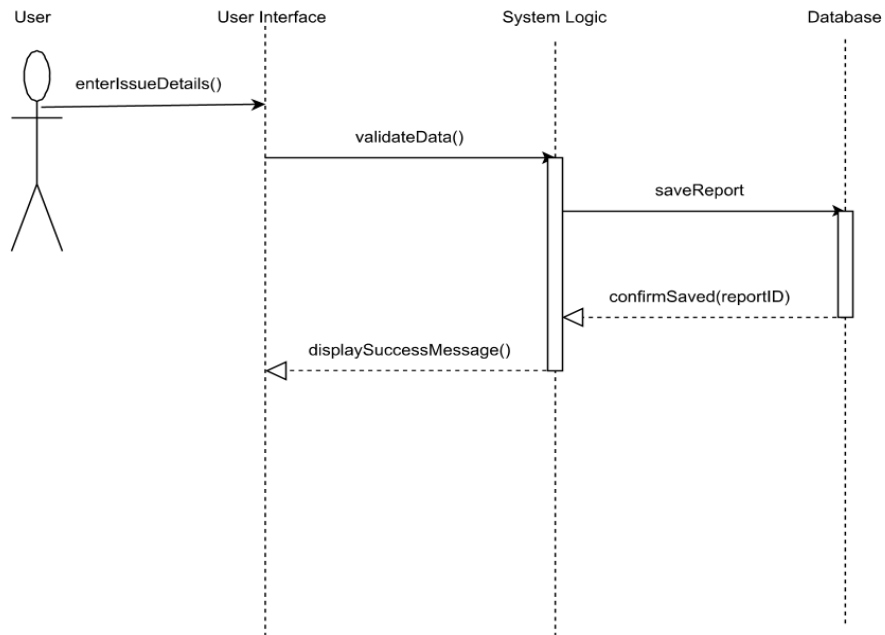


3.4. Class Diagram



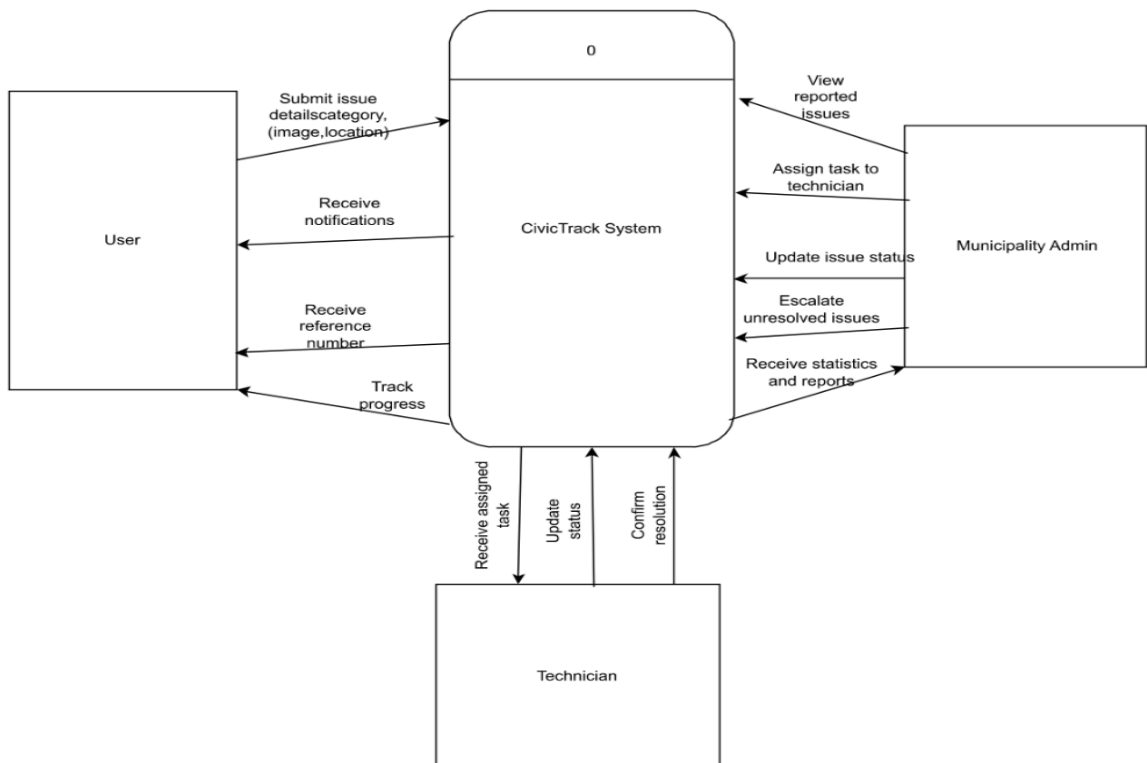
3.5. Sequence Diagram

Sequence Diagram for issue submission



3.6. Context Diagram

Context Diagram



4. Project Management Artefacts

4.1 Work Breakdown Structure (WBS)

The table below decomposes the **Analysis Stage** and **Detailed System Blueprint** into specific tasks, deliverables, and assignments to ensure project milestones are met.

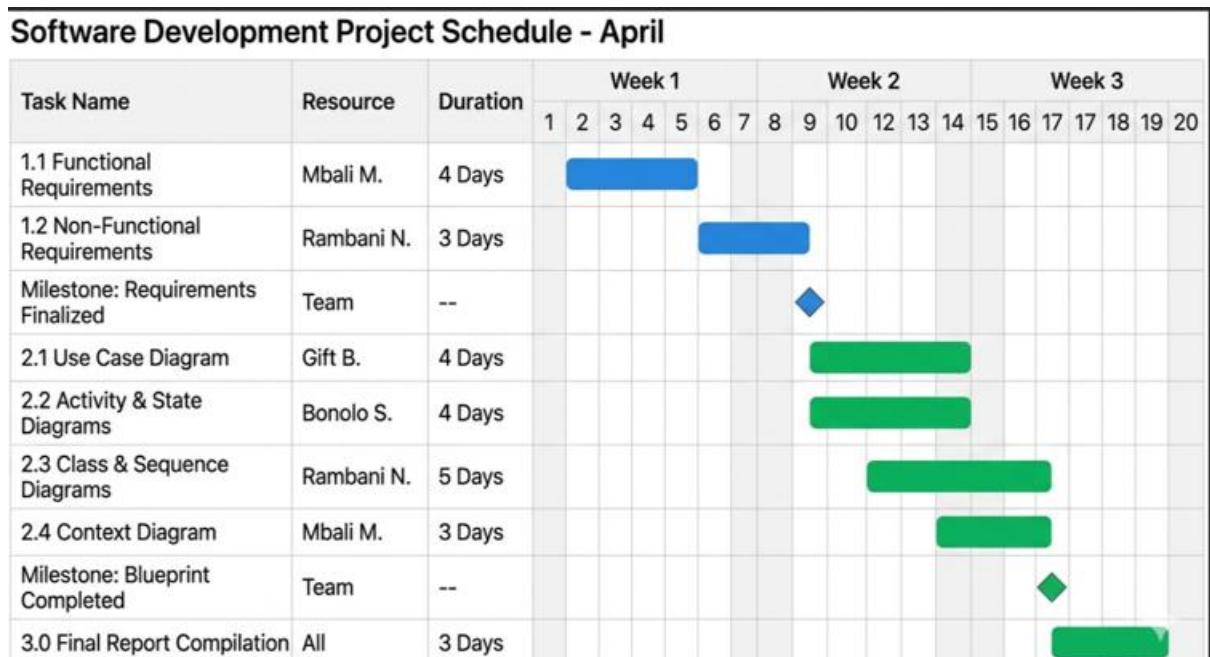
Task ID	Task Description	Deliverables	Resources Allocated	Dependencies	Milestones
1.0	Requirement Definition	Validated Specs	All Members	None	Baseline Set
1.1	Draft Functional Requirements	Input/Process/Output List	Mbali Innocent Msiza	None	Draft Ready
1.2	Define Non-Functional Requirements	Quality Attribute Table	Rambani Ntendele ni	None	Draft Ready
2.0	System Blueprinting	UML Diagram Set	Technical Team	1.0	Design Locked
2.1	Create Use Case Diagram	Use Case Visual	Gift Baloyi	1.1	Visual Verified
2.2	Design Activity &	Process Flow Visuals	Bonolo Sesinyi	1.1	Flows Ready
2.3	Class & Sequence Diagrams	Data/Logic Visuals	Rambani Ntendele ni	2.1	Logic Validated
2.4	Assemble Context Diagram	System Scope Visual	Mbali Innocent Msiza	1.1	Scope Defined
3.0	System Blueprinting	Phase 2 Report	All Members	2.0	Submission

4.2 Risk Management and Contingency Planning

Identified Risk	Contingency Plan
Logic Mismatch: Differences between text requirements and UML diagram logic.	Conduct a crosswalk audit to ensure every functional requirement is represented in at least one system diagram.
Technical Bottlenecks: Delays in mapping complex logic for Sequence and Class diagrams.	Establish a peer-review system between technical leads to troubleshoot logic flows early.
Version Control: Potential for data loss or conflicting edits during team collaboration.	Mandate the use of a shared cloud-based repository with automated version history and "Track Changes" enabled.
Scope Creep: System boundaries in the Context Diagram exceeding the project brief.	Re-verify all external actors (Citizen, Admin, Technician) against the initial scope before finalizing the blueprint.

4.3 Gantt Chart

The Gantt Chart provides a visual timeline of the task durations and dependencies mapped in the WBS



References

JGraph (2026) *draw.io* (Version 24.7.5) [Software]. Available at: <https://app.diagrams.net/> (Accessed: 20 April 2026).

Kisten, M. (2026) 'NPRT63 Project (ICT 3rd Year): Lecture 2' [Lecture to ICT 3rd Year Students], *NPRT63*, Sol Plaatje University, 3 March.